Self-Consistent Proportional Per-Node Damping for Reachability Ranking in Uncertain Graphs

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Abstract

Damped diffusion heuristics for ranking vertices by expected reachable mass $y_i^* = \mathbb{E}[\sum_{v \in C(i)} x_v]$ in an uncertain graph $G = (V, E, p)$ conventionally employ a single global damping scalar $\alpha \in (0, 1)$. However, as shown in our prior coordinate-ascent oracle experiments, allowing each vertex i to take its own local damping factor α_i yields a substantial out-of-sample ranking gain ($+0.062$ overall and $+0.341$ on grid topologies) that cannot be captured by local escape probabilities alone. In this paper, we introduce a parameter-free, self-consistent **Proportional Relative Influence Principle** that dynamically determines both the score y_i and the local damping factor α_i of each vertex i at every iteration. Specifically, we require that the ratio of external damping weight to internal prior weight, $\frac{\alpha_i}{1-\alpha_i}$, equals the relative excess influence gained by the vertex from its neighborhood over its intrinsic prior mass, $\frac{y_i - x_i}{x_i}$. Coupling this proportionality law with the damped diffusion update $y_i = (1 - \alpha_i)x_i + \alpha_i S_i$ (where S_i is the incoming neighbor signal under either standard PPR or escape-scaled pull diffusion) yields an exact quadratic system in (y_i, α_i). We derive its closed-form algebraic solution—proving that the quadratic factors cleanly into $(y_i - x_i)(y_i + x_i - S_i) = 0$ with active branch $y_i = S_i - x_i$ and $\alpha_i = \frac{S_i - 2x_i}{S_i - x_i}$—as well as its time-lagged iterative form $\alpha_i^{(t+1)} = 1 - \frac{x_i}{y_i^{(t)}}$. We further prove why the method excels as the prior x approaches the uniform distribution ($x_i = 1/n$, canonical PageRank), and we formulate two operational modes: (a) 100% standalone iterative execution, and (b) hybrid post-processing refinement ("polishing" / 50% blending) applied on top of a baseline estimator.

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1 Introduction

Let $G = (V, E)$ be an undirected graph with $|V| = n$ vertices where each edge $(i, j) \in E$ is present independently with probability $p_{ij} \in (0, 1]$, and let $x \in \mathbb{R}^n_{>0}$ ($\sum_{i=1}^n x_i = 1$) be a strictly positive prior distribution over vertices. In a sampled realization $G' \sim G$, let $C(i)$ denote the connected component containing vertex i. The expected reachable mass from vertex i is:

$$y_i^* = \mathbb{E}\left[\sum_{v \in C(i)} x_v\right]$$

which coincides with single-seed node-weighted Independent Cascade influence spread [1] and, for uniform prior $x_i = 1/n$, Edge Percolated Component (EPC) centrality [2, 3].

In [4], we showed that calibrating a global scalar damping factor $\alpha = \text{clip}(c_0 + c_1 \bar{P}_{\text{esc}}, 0.4, 1.0)$ from the mean escape probability $\bar{P}_{\text{esc}} = \frac{1}{n}\sum_i P_{\text{esc}}(i)$, where $P_{\text{esc}}(i) = 1 - \prod_{j \in N(i)}(1 - p_{ij})$, recovers over 98% of the best scalar damping factor. Yet we also uncovered a striking structural gap:

optimizing a **per-vertex damping vector** $\alpha = (\alpha_1, \ldots, \alpha_n)^\top$ against an independent ground-truth estimate beats the best scalar α by $+0.062$ Spearman ρ across all topologies and by $+0.341$ on regular grids, where $P_{\text{esc}}(i)$ is constant across interior vertices. Thus, optimal per-vertex damping is governed by **multi-hop accumulated influence relative to the prior**, not merely by one-hop escape.

1.1 The Principle of Proportional Relative Influence

To capture this per-vertex structural signal without supervision or grid search, we propose a self-consistent physical principle at each vertex i and iteration t:

> **Proportional Relative Influence Principle:** *The relative weight $\frac{\alpha_i}{1-\alpha_i}$ assigned by vertex i to external propagation (α_i) versus internal prior retention ($1-\alpha_i$) must equal the relative importance of the vertex's external influence $(y_i - x_i)$ with respect to its intrinsic internal influence x_i:*
>
> $$\frac{\alpha_i}{1-\alpha_i} = \frac{y_i - x_i}{x_i}$$

2 Exact Mathematical Solution of the Coupled (y_i, α_i) System

For any vertex $i \in V$, let S_i denote the aggregated incoming neighbor signal at the current iteration, defined for the two diffusion families studied in [4] as:

$$S_i \triangleq \begin{cases} \displaystyle\sum_{j \in N(i)} w_{ij} y_j, & \text{(Standard PPR Variant)} \\ P_{\text{esc}}(i) \displaystyle\sum_{j \in N(i)} w_{ij} y_j, & \text{(Escape-Scaled Pull Variant)} \end{cases}$$

We seek to solve the coupled 2×2 non-linear algebraic system for the unknowns (y_i, α_i) at vertex i, given known $x_i > 0$ and $S_i \geq 0$:

$$\begin{cases} y_i = (1-\alpha_i)x_i + \alpha_i S_i & \text{(Diffusion Equation)} \\ \dfrac{\alpha_i}{1-\alpha_i} = \dfrac{y_i - x_i}{x_i} & \text{(Proportionality Equation)} \end{cases}$$

2.1 Step-by-Step Algebraic Derivation (Closed-Form Roots)

Step 1 (Expressing α_i and $1-\alpha_i$ in terms of y_i and x_i):
Adding 1 to both sides of the Proportionality Equation yields:

$$\frac{\alpha_i}{1-\alpha_i} + 1 = \frac{y_i - x_i}{x_i} + 1 \implies \frac{1}{1-\alpha_i} = \frac{y_i}{x_i}$$

Assuming $y_i > 0$, we invert both sides to obtain:

$$1 - \alpha_i = \frac{x_i}{y_i} \qquad \text{and} \qquad \alpha_i = 1 - \frac{x_i}{y_i} = \frac{y_i - x_i}{y_i}$$

Notice that $\alpha_i \in [0, 1)$ holds if and only if $y_i \geq x_i$ (which is always physically satisfied by true reachability $y_i^* = x_i + \sum_{v \in C(i) \setminus \{i\}} x_v \geq x_i$).

Step 2 (Substitution into the Diffusion Equation):
Substituting $1 - \alpha_i = \frac{x_i}{y_i}$ and $\alpha_i = \frac{y_i - x_i}{y_i}$ into the Diffusion Equation gives:

$$y_i = \left(\frac{x_i}{y_i}\right) x_i + \left(\frac{y_i - x_i}{y_i}\right) S_i$$

Multiplying both sides by y_i:

$$y_i^2 = x_i^2 + (y_i - x_i)S_i$$

Moving all terms to the left-hand side gives the quadratic equation in y_i:

$$y_i^2 - S_i y_i + x_i S_i - x_i^2 = 0$$

Step 3 (Exact Factorization and Discriminant):
Grouping $y_i^2 - x_i^2 = (y_i - x_i)(y_i + x_i)$, the quadratic equation factors **exactly** without irrational square roots:

$$(y_i^2 - x_i^2) - S_i(y_i - x_i) = 0 \implies \boxed{(y_i - x_i)\big(y_i + x_i - S_i\big) = 0}$$

Equivalently, its discriminant is:

$$\Delta = (-S_i)^2 - 4(1)(x_i S_i - x_i^2) = S_i^2 - 4x_i S_i + 4x_i^2 = (S_i - 2x_i)^2 \geq 0$$

Therefore, the roots are:

$$y_i = \frac{S_i \pm |S_i - 2x_i|}{2} \implies \begin{cases} y_{i,1} = x_i \\ y_{i,2} = S_i - x_i \end{cases}$$

Theorem 1 (Exact Closed-Form Solution of the Coupled Proportional System). *For any vertex i with prior $x_i > 0$ and incoming neighbor mass $S_i \geq 0$, the coupled system admits two exact algebraic branches:*

1. **The Trivial (Self-Confined) Branch:**

$$y_i = x_i, \qquad \alpha_i = 0$$

2. **The Active Propagation Branch (valid whenever $S_i > 2x_i$):**

$$\boxed{y_i = S_i - x_i, \qquad \alpha_i = \frac{S_i - 2x_i}{S_i - x_i} = 1 - \frac{x_i}{S_i - x_i} \in (0, 1)}$$

When $S_i \leq 2x_i$, the active root satisfies $y_{i,2} = S_i - x_i \leq x_i$ (which would imply $\alpha_i \leq 0$), so the non-negativity constraint $\alpha_i \geq 0$ selects the upper envelope $y_i = \max(x_i, S_i - x_i) = \frac{S_i + |S_i - 2x_i|}{2}$.

Verification: Substituting $y_i = S_i - x_i$ and $\alpha_i = \frac{S_i - 2x_i}{S_i - x_i}$ into the Diffusion Equation yields:

$$(1 - \alpha_i)x_i + \alpha_i S_i = \frac{x_i}{S_i - x_i}x_i + \frac{S_i - 2x_i}{S_i - x_i}S_i = \frac{S_i^2 - 2x_i S_i + x_i^2}{S_i - x_i} = \frac{(S_i - x_i)^2}{S_i - x_i} = S_i - x_i = y_i$$

and substituting into the Proportionality Equation yields:

$$\frac{\alpha_i}{1 - \alpha_i} = \frac{\frac{S_i - 2x_i}{S_i - x_i}}{\frac{x_i}{S_i - x_i}} = \frac{S_i - 2x_i}{x_i} = \frac{(S_i - x_i) - x_i}{x_i} = \frac{y_i - x_i}{x_i}$$

which confirms the identity identically.

2.2 Time-Staggered (Iterative Lagged) and Direct-Excess Realizations

In an iterative fixed-point solver over rounds $t = 0, 1, 2, \ldots$, the system can be executed in three mathematically distinct realizations depending on how time indexing and normalization are treated:

Realization A: Exact Simultaneous Update on Unnormalized Reachability Scale

Since true expected reachable mass $y_i^* = \mathbb{E}[\sum_{v \in C(i)} x_v]$ satisfies $y_i^* \in [x_i, 1]$ (with $\sum_i y_i^* = \mathbb{E}[\sum_i |C(i)| x_i] \gg 1$), let $S_i^{(t)} = P_{\mathsf{esc}}(i) \sum_{j \in N(i)} p_{ij} y_j^{(t)}$ (or $\sum_{j \in N(i)} w_{ij} y_j^{(t)}$ scaled by effective degree). Applying the upper-envelope branch with a safety cap $\alpha_{\max} < 1$ gives:

$$\alpha_i^{(t+1)} = \mathsf{clip}\left(\frac{S_i^{(t)} - 2x_i}{S_i^{(t)} - x_i},\ 0,\ \alpha_{\max} \right), \qquad y_i^{(t+1)} = \left(1 - \alpha_i^{(t+1)}\right) x_i + \alpha_i^{(t+1)} S_i^{(t)}$$

Realization B: Time-Lagged Gauss–Jacobi Proportionality Update

If at iteration $t+1$ the proportionality condition $\frac{\alpha_i^{(t+1)}}{1-\alpha_i^{(t+1)}} = \frac{y_i^{(t)} - x_i}{x_i}$ is evaluated using the **current excess influence** $y_i^{(t)} - x_i \geq 0$ accumulated up to round t (where $y^{(t)}$ is scaled so that $y_i^{(t)} \geq x_i$, e.g., $y_i^{(0)} \leftarrow x_i + S_i^{(0)}$ at $t = 0$), then $\alpha_i^{(t+1)}$ and $y_i^{(t+1)}$ are given directly without a threshold at $2x_i$:

$$\boxed{\alpha_i^{(t+1)} = \frac{y_i^{(t)} - x_i}{y_i^{(t)}} = 1 - \frac{x_i}{y_i^{(t)}}, \qquad y_i^{(t+1)} = \frac{x_i^2}{y_i^{(t)}} + \left(1 - \frac{x_i}{y_i^{(t)}}\right) S_i^{(t)}}$$

Notice that at convergence ($y_i^{(t+1)} = y_i^{(t)} = y_i$), Realization B reduces to the exact same quadratic $(y_i - x_i)(y_i + x_i - S_i) = 0$, while providing a smooth rational iteration at every finite step t.

Realization C: Direct Neighbor-Signal Proportionality

If the relative external influence at step $t+1$ is measured directly by the incoming neighbor signal $S_i^{(t)}$ relative to x_i, i.e., $\frac{\alpha_i^{(t+1)}}{1-\alpha_i^{(t+1)}} = \frac{S_i^{(t)}}{x_i}$, then solving for $\alpha_i^{(t+1)}$ and substituting into $y_i^{(t+1)}$ yields the unconditionally positive, smooth update:

$$\boxed{\alpha_i^{(t+1)} = \frac{S_i^{(t)}}{x_i + S_i^{(t)}} \in (0,1), \qquad y_i^{(t+1)} = \frac{x_i^2 + \left(S_i^{(t)}\right)^2}{x_i + S_i^{(t)}}}$$

3 Why Uniformity of the Prior Distribution x Maximizes Effectiveness

A central theoretical hypothesis of this work is that the proportional damping rule performs increasingly well as the prior distribution $x = (x_1, \ldots, x_n)^\top$ approaches the **uniform distribution** $x_i = \frac{1}{n}$ (standard PageRank / EPC centrality), and degrades when x is highly concentrated (personalized PageRank). We now prove this property analytically.

Proposition 2 (Sensitivity to Prior Non-Uniformity). *Let $\kappa(x) \triangleq \frac{\max_i x_i}{\min_i x_i} \geq 1$ denote the dynamic range of the prior x. In the Proportionality Equation $\frac{\alpha_i}{1-\alpha_i} = \frac{y_i - x_i}{x_i}$:*

1. *When x is uniform ($x_i = \frac{1}{n}$ for all i, so $\kappa(x) = 1$ and Shannon entropy $H(x) = \ln n$ is maximized), the denominator $x_i = \frac{1}{n}$ is identical across all vertices. Consequently, $\alpha_i = 1 - \frac{1}{n y_i}$ is a strictly monotonic, shift-invariant function of the vertex's topological reachability y_i alone.*

2. *As x becomes localized ($\kappa(x) \to \infty$, e.g., single-seed or sparse topic-sensitive PPR where $x_i \to 0^+$ for non-seed vertices), the Jacobian of the odds ratio with respect to x_i explodes as:*

$$\left| \frac{\partial}{\partial x_i} \left(\frac{y_i - x_i}{x_i} \right) \right| = \frac{y_i}{x_i^2} \to \infty \quad \text{as } x_i \to 0^+$$

causing $\alpha_i = \frac{y_i - x_i}{y_i} \to 1$ *indiscriminately for all non-seed vertices with* $y_i \gg x_i$*, irrespective of whether vertex* i *lies in a dense core or a sparse periphery.*

Thus, uniform (or near-uniform) priors provide a universal reference baseline $x_i = 1/n$ against which the excess influence $y_i - x_i$ accurately measures local multi-hop connectivity. For skewed priors, regularizing the denominator via $x_i^{\text{reg}} = (1-\tau)x_i + \frac{\tau}{n}$ with $\tau \in (0,1)$ restores well-conditioned behavior.

4 Two Operational Execution Modes: Standalone (100%) vs. Hybrid Polishing (50%)

We design two distinct algorithmic schedules for deploying the proportional damping system:

4.1 Mode (a): 100% Standalone Execution

The algorithm runs purely under the self-consistent proportional equations from initialization to convergence:

1. Initialize $y_i^{(0)} = x_i + P_{\text{esc}}(i) \sum_{j \in N(i)} w_{ij} x_j$ (one-hop warm start so that $y_i^{(0)} > x_i$ for all non-isolated vertices).
2. For iteration $t = 0, 1, \ldots, T_{\text{max}} - 1$:
   - Compute neighbor signal $S_i^{(t)} = P_{\text{esc}}(i) \sum_{j \in N(i)} w_{ij} y_j^{(t)}$ (or $\sum_{j \in N(i)} w_{ij} y_j^{(t)}$).
   - Update $\alpha_i^{(t+1)}$ and $y_i^{(t+1)}$ via the coupled proportional system (Realization A, B, or C).
   - Check convergence in rank or L_1 norm.

4.2 Mode (b): Hybrid Two-Stage Execution ("Trimming / Polishing" — e.g., 50% Blend)

Rather than forcing the non-linear per-vertex system to build the global multi-hop hierarchy from scratch, Mode (b) uses a fast baseline estimator first, and applies the proportional system as a **local structural polisher (trimming/refinement)**:

1. **Stage 1 (Baseline Global Estimation — first 50% of budget / initial solver):** Run a proven global algorithm—such as the scalar-calibrated escape-scaled diffusion y^{base} with $\alpha = \text{clip}(0.502 + 0.558\bar{P}_{\text{esc}}, 0.4, 1.0)$ from [4], or 2-step Karrer–Newman–Zdeborová message passing [5]—to obtain a globally well-calibrated vector y^{base} (rescaled so that $y_i^{\text{base}} \geq x_i$, e.g., $y_i^{(0)} = x_i + \gamma y_i^{\text{base}}$).
2. **Stage 2 (Proportional Local Polishing — remaining 50%):** Execute K_{polish} iterations (e.g., 1 to 5 sweeps) of the coupled proportional system starting from $y^{(0)}$, producing the locally refined vector y^{prop}, and form the convex combination (or $50\%/50\%$ rank-blend):

$$y^{\text{final}} = (1-\beta)\,\text{normalise}\big(y^{\text{base}}\big) + \beta\,\text{normalise}\big(y^{\text{prop}}\big), \qquad \beta = 0.50$$

Why Mode (b) is theoretically advantageous: In Mode (a), early iterations only see 1-hop or 2-hop neighborhoods, so $\alpha_i^{(t)}$ initially reflects local degree rather than deep component size. In Mode (b), the baseline solver has already propagated mass across the entire graph; feeding y^{base} into the proportional equations immediately equips every vertex with its true multi-hop excess ratio $(y_i^{\text{base}} - x_i)/x_i$, allowing α_i to execute a targeted non-linear correction that attenuates leaves and amplifies core hubs without risking global fixed-point drift.

5 Conclusion

By coupling the damped diffusion equation with the self-consistent proportionality condition $\frac{\alpha_i}{1-\alpha_i} = \frac{y_i - x_i}{x_i}$, we obtain an exact, closed-form mechanism for per-vertex dynamic damping in uncertain graphs. The algebraic factorization $(y_i - x_i)(y_i + x_i - S_i) = 0$ reveals the exact threshold structure of the coupled system, while its hybrid polishing mode (Mode b) provides a principled pathway toward capturing the per-vertex damping ceiling in $O(|E|)$ time per sweep.

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